

Report

“Cascading Bandits : Learning to Rank in the Cascade Model”

Branislav Kveton, Csaba Szepesvari, Zheng Wen , Azin Ashkan (2015)

Report prepared by: Swarnim Verma

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1. Overview of the Paper

1.1. Problem Statement

We consider the problem of learning to rank in the setting known as cascade model, where a user is presented with a list of items and examines them sequentially until they find an attractive one. The goal is to learn which items are most attractive to users based on their interactions with the presented lists. This problem is relevant in applications such as web search and recommendation systems, where the order of items can significantly impact user engagement. Formally, suppose for a particular query search engine has L web pages to recommend and it can only recommend K web pages at a time. The user examines the recommended web pages in order and clicks on the first attractive web page they encounter, if any. The attractiveness of each web page(it depends on the text or images shown as preview) is unknown and must be learned from user interactions. The challenge is to balance exploration (learning about the attractiveness of different items) and exploitation (recommending items that are currently believed to be attractive) in order to maximize user satisfaction over time.

1.2. Key Idea

The key idea of the paper is to model the learning to rank problem in the cascade model as a multi-armed bandit problem. The authors propose two algorithms, CascadeUCB1 and CascadeKL-UCB, which are adaptations of the UCB1 and KL-UCB algorithms for this setting. These algorithms maintain confidence bounds on the attractiveness of each item and use these bounds to select which items to recommend at each time step.

1.3. Contributions

The main contributions of this paper are:

- **Contribution 1:** propose two UCB-based algorithms, *CascadeUCB1* and *CascadeKL-UCB*, for learning to rank in the cascade model.
- **Contribution 2:** derive gap-dependent upper bounds on the regret of the proposed algorithms.
- **Contribution 3:** establish a lower bound on the regret for a specific class of problem instances.

- **Contribution 4:** conduct synthetic experiments and compare two variants (ascending UCB and descending UCB) of both proposed algorithms.
- **Contribution 5:** compare our proposed methods with ranked bandits algorithm in the DBN model, demonstrating that our algorithms perform well even when the cascade model assumptions are violated.

2. Previous Work

2.1. Previous Work

- **An Experimental Comparison of Click Position-Bias Models [1]:** This paper compares several hypothesis about how positional bias occurs(suppose two same webpages or webpages having same attractiveness(though attractiveness may sound vague will be formalised later) are placed at different positions but still the one which appears first is clicked more number of times). Few hypothesis like baseline, Mixture, examination hypothesis and cascade model are compared and it is found that cascade model is a reasonable fit for click data obtained from real life user feedbacks from search engines.
- **UCB1 and KL_UCB [6, 7]:** These are two well-known algorithms for the multi-armed bandit problem. UCB1 maintains an upper confidence bound for each arm based on the number of times it has been pulled and the rewards received, while KL_UCB uses Kullback-Leibler divergence to compute tighter confidence bounds. Both algorithms achieve logarithmic regret in the standard multi-armed bandit setting, but they do not directly address the specific challenges of learning to rank in the cascade model. Both the proposed algorithms in this paper are adaptations of these two algorithms to the cascade model setting.
- **DBN model and ranked bandits [2, 5]:** The Dynamic Bayesian Network (DBN) model is another even more sophisticated approach to model user interactions with ranked lists, where the user has a probability of being satisfied with each item and may continue to examine the next item with some probability even after clicking the item or abandon the search altogether. The ranked bandits algorithm is designed for the DBN model and maintains separate bandit algorithms for each position in the ranked list also each of those bandit algorithms have KL-UCB as the base algorithm.

3. What is Cascade Model Formally?

In this model, the user scans a list of K items $A = (a_1, \dots, a_K) \in \Pi_K(E)$ from the first item a_1 to the last a_K , where $\Pi_K(E)$ is the set of all K -permutations of set E . The model is parameterized by mean attraction probabilities $\bar{w} \in [0, 1]^E$. After the user examines item a_k , the item attracts the user with probability $\bar{w}(a_k)$, independently of the other items. If the user is attracted by item a_k , the user clicks on it and does not examine the remaining items. If the user is not attracted by item a_k , the user examines item a_{k+1} . It is easy to see that the probability that item a_k is examined is $\prod_{i=1}^{k-1} (1 - \bar{w}(a_i))$, and that the probability that at least one item in A is attractive is $1 - \prod_{i=1}^K (1 - \bar{w}(a_i))$. This objective is maximized by K most attractive items. The cascade model assumes that the user clicks on at most one item

4. Generalised Cascading Bandits : Learning the intrinsic parameters of the cascade model

The goal of this paper is to learn the attraction probabilities $\bar{w}$ from user interactions. The learning agent interacts with the user in a sequence of T rounds. In round t , the agent recommends a list of K items $A_t \in \Pi_K(E)$ to the user. The user examines the list and clicks on at most one item according to the cascade model. The agent observes the index of the clicked item, if any, and receives a reward of 1 if the user clicks on an item and 0 otherwise.

5. Mathematical Formulation of the Problem

Definition 5.1. We represent the problem instance by a tuple $\mathbf{B} = (\mathbf{E}, \mathbf{P}, \mathbf{K})$, where $E = \{1, \dots, L\}$ is a set of L items, P is a discrete probability distribution over a unit hypercube $\{0, 1\}^E$, and $K \leq L$ is the number of recommended items. We call this **generalized bandit** because the form of the distribution P has not been specified yet.

Let $(w_t)_{t=1}^T$ be an i.i.d. sequence of T weights drawn from P , where $w_t \in \{0, 1\}^E$ and $w_t(e)$ is the preference of the user for item e at time t . That is, $w_t(e) = 1$ if and only if item e attracts the user at time t . The learning agent interacts with our problem as follows. At time t , the agent recommends a list of K items $A_t = (a_1^t, \dots, a_K^t) \in \Pi_K(E)$. The list is computed from the observations of the agent up to time t . The user examines the list, from the first item a_1^t to the last a_K^t , and clicks on the first attractive item. If the user is not attracted by any item, the user does not click on any item. Then time increases to $t + 1$. The reward of the agent at time t can be written in several forms. For instance, as $\max_k w_t(a_k^t)$, at least one item in list A_t is attractive; or as $f(A_t, w_t)$, where:

$$f(A, w) = 1 - \prod_{k=1}^K (1 - w(a_k)),$$

$A = (a_1, \dots, a_K) \in \Pi_K(E)$, and $w \in \{0, 1\}^E$. This later algebraic form is particularly useful in our proofs. The agent at time t receives feedback:

$$C_t = \arg \min \{1 \leq k \leq K : w_t(a_k^t) = 1\},$$

where we assume that $\arg \min \emptyset = \infty$. The feedback C_t is the click of the user. If $C_t \leq K$, the user clicks on item C_t . If $C_t = \infty$, the user does not click on any item. Since the user clicks on the first attractive item in the list, we can determine the observed weights of all recommended items at time t from C_t . In particular, note that:

$$w_t(a_k^t) = \mathbf{1}\{C_t = k\} \quad k = 1, \dots, \min\{C_t, K\}. \quad (1)$$

We say that item e is observed at time t if $e = a_k^t$ for some $1 \leq k \leq \min\{C_t, K\}$.

6. Cascading Bandits : Solves a Specific Class of Problems

Definition 6.1. We will assume for simplicity that the weights w_t that are drawn from a distribution P look like $[w_t(1), w_t(2), \dots, w_t(L)]$ where $w_t(e) \sim \text{Bernoulli}(\bar{w}(e))$. We call this class of problems as **cascading bandits**.

Definition 6.2. Later we will consider even simpler class of problems where aside from above assumption we have $\bar{w}(e) = p$ for $e \leq K$ and $\bar{w}(e) = p - \Delta$ for $e > K$ for some $p \in (0, 1)$ and $\Delta \in (0, p)$. We will denote an instance of such problems by $\mathbf{B}_{\text{easy}}(\mathbf{L}, \mathbf{K}, \mathbf{p}, \Delta)$.

Definition 6.3. We say that the algorithm is **consistent** if for any cascading bandit instance, any suboptimal list A , and any $\alpha > 0$, $\mathbb{E}[N_T(A)] = o(T^\alpha)$, where $N_T(A)$ is the number of times that list A is recommended in T steps. Note that the restriction to the consistent algorithms is without loss of generality. The reason is that any inconsistent algorithm must suffer polynomial regret on some instance of cascading bandits, and therefore cannot achieve logarithmic regret on every instance of our problem.

6.1. Regret

The expected reward for list $A \in \Pi_K(E)$, the probability that at least one item in A is attractive, can be expressed as $\mathbb{E}[f(A, w)] = f(A, \bar{w})$ due to form of P , and depends only on the attraction probabilities of individual items in A .

Definition 6.4. The agent's policy is evaluated by its expected cumulative regret:

$$R(n) = \mathbb{E} \left[\sum_{t=1}^T R(A_t, w_t) \right],$$

where $R(A_t, w_t) = f(A^*, w_t) - f(A_t, w_t)$ is the regret of the agent at time t and:

$$A^* = \arg \max_{A \in \Pi_K(E)} f(A, \bar{w})$$

is the optimal list of items, the list that maximized the reward at any time t .

Since f is invariant to the permutation of A , there exist at least K optimal lists. WLOG we assume $\bar{w}(1) \geq \dots \geq \bar{w}(L)$ and $A^* = (1, \dots, K)$. We say that item e is optimal if $e \leq K$ and suboptimal otherwise. We also define the gap between a item e and the item e^* as $\Delta_{e,e^*} = \bar{w}(e^*) - \bar{w}(e)$.

We introduce a permutaion π_t on optimal items ie. $e = 1$ to $e = K$ to describe the event $G_{e,e^*,t}$. For any k , if the k -th item in A_t is optimal, we place this item at position k : $\pi_t(k) = a_k^t$. The remaining optimal items are positioned arbitrarily.

$$G_{e,e^*,t} = \left\{ \exists 1 \leq k \leq K \text{ s.t. } a_k^t = e, \pi_t(k) = e^*, w_t(a_1^t) = \dots = w_t(a_{k-1}^t) = 0 \right\}$$

This event is the key to our analysis. This event signifies two things depending on whether e is optimal or suboptimal. If e is suboptimal, it signifies a bad event(in the sense that it misleads the learning agent) as follows : The first condition ensures that suboptimal item e is in the recommended list A_t . The second condition ensures that optimal item e^* is not in the recommended list A_t because if it were, it would be placed at position $j \neq k$ and hence $\pi_t(j) = e^*$ but we have been given $\pi_t(k) = e^*$. The third condition ensures that item e is observed at time t because if any of the items before position k were attractive (i.e., had a weight of 1), the user would have clicked on one of those items and not observed item e .

If e is optimal, it signifies that an optimal item was observed at time t at some position k .

Since A^* is optimal with respect to $\bar{w}$, $\bar{w}(a_k^t) \leq \bar{w}(\pi_t(k))$ for all k . Similarly, since A_t is optimal with respect to U_t , $U_t(a_k^t) \geq U_t(\pi_t(k))$ for all k . We will use this observation in the proof of the main results later.

Remark 1. The regret incurred in any step is not affected by the order of recommended items. However the order of recommended items does decide the factor in regret bound.(We will see the experimental results later.)

7. Description of the Algorithms

7.1. Algorithm 1: [CascadeUCB1]

Algorithm 1 CascadeUCB1

Require: number of arms to be recommended K out of L total arms, horizon T

- 1: sample $w_0 \sim P$ and Initialize $N_0(e) \leftarrow 1$ and $\hat{w}_1(e) \leftarrow w_0(e)$ for all $e \in E$
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: Compute UCBs $U_t(e)$ for all $e \in E$ where $U_t(e) = \hat{w}_{N_{t-1}(e)}(e) + \sqrt{\max\left(0, \frac{3 \log(t-1)}{2N_{t-1}(e)}\right)}$
 - 4: Recommend $A_t = (a_1^t, \dots, a_K^t)$ where $a_1^t, \dots, a_K^t$ are the K items with largest $U_t(e)$
 - 5: Observe click $C_t \in \{1, \dots, K, \infty\}$
 - 6: $\forall e \in E$, set $N_t(e) \leftarrow N_{t-1}(e)$
 - 7: **for** $k = 1, \dots, \min\{C_t, K\}$ **do**
 - 8: $e \leftarrow a_k^t$
 - 9: $N_t(e) \leftarrow N_{t-1}(e) + 1$
 - 10: $\hat{w}_{N_t(e)}(e) \leftarrow \frac{N_{t-1}(e) \hat{w}_{N_{t-1}(e)}(e) + \mathbf{1}\{C_t=k\}}{N_t(e)}$ $\triangleright$ This lowers the empirical weight of items that were observed but not clicked and increases the same for item that was clicked.
 - 11: **end for**
 - 12: **end for**
 - 13: **return** estimated best K arms
-

7.2. Algorithm 2: [CascadeKL-UCB]

Algorithm 2 CascadeKL-UCB

Require: number of arms to be recommended K out of L total arms, horizon T

- 1: sample $w_0 \sim P$ and Initialize $N_0(e) \leftarrow 1$ and $\hat{w}_1(e) \leftarrow w_0(e)$ for all $e \in E$
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: Compute UCBs $U_t(e)$ for all $e \in E$ where

$$U_t(e) = \max\left\{q \in [\hat{w}_{N_{t-1}(e)}(e), 1] : N_{t-1}(e) D_{\text{KL}}(\hat{w}_{N_{t-1}(e)}(e) \parallel q) \leq \log t + 3 \log \log t\right\}$$
 - 4: Recommend $A_t = (a_1^t, \dots, a_K^t)$ where $a_1^t, \dots, a_K^t$ are the K items with largest $U_t(e)$
 - 5: Observe click $C_t \in \{1, \dots, K, \infty\}$
 - 6: $\forall e \in E$, set $N_t(e) \leftarrow N_{t-1}(e)$
 - 7: **for** $k = 1, \dots, \min\{C_t, K\}$ **do**
 - 8: $e \leftarrow a_k^t$
 - 9: $N_t(e) \leftarrow N_{t-1}(e) + 1$
 - 10: $\hat{w}_{N_t(e)}(e) \leftarrow \frac{N_{t-1}(e) \hat{w}_{N_{t-1}(e)}(e) + \mathbf{1}\{C_t=k\}}{N_t(e)}$ $\triangleright$ This lowers the empirical weight of items that were observed but not clicked and increases the same for item that was clicked.
 - 11: **end for**
 - 12: **end for**
 - 13: **return** estimated best K arms
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7.3. Algorithm 3: RankedKL-UCB

Algorithm 3 RankedKL-UCB

Require: Total items L , list length K , horizon T

- 1: Initialize $N_0(k, e) \leftarrow 1$ and $\hat{w}(k, e) \leftarrow 1[\text{rand}() < \bar{w}(e)]$ for all $k \in \{1, \dots, K\}, e \in E$
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: $A_t \leftarrow []$, chosen $\leftarrow \emptyset$
- 4: **for** $k = 1, \dots, K$ **do**
- 5: Compute KL-UCB for all unchosen items:

$$U_t(k, e) = \max \{q \in [\hat{w}(k, e), 1] : N_{t-1}(k, e) D_{\text{KL}}(\hat{w}(k, e) \parallel q) \leq \log t + 3 \log \log t\}$$

- 6: $a_k^t \leftarrow \arg \max_{e \notin \text{chosen}} U_t(k, e)$
 - 7: Append a_k^t to A_t , add a_k^t to chosen
 - 8: **end for**
 - 9: Recommend list $A_t = (a_1^t, \dots, a_K^t)$, observe click C_t
 - 10: **for** $k = 1, \dots, \min(C_t, K)$ **do**
 - 11: $e \leftarrow a_k^t$
 - 12: $N_t(k, e) \leftarrow N_{t-1}(k, e) + 1$
 - 13: $\hat{w}(k, e) \leftarrow \frac{N_{t-1}(k, e) \hat{w}(k, e) + \mathbf{1}\{C_t=k\}}{N_t(k, e)} \triangleright$ This lowers the empirical weight of items that were observed but not clicked and increases the same for item that was clicked.
 - 14: **end for**
 - 15: **end for**
-

8. Main Results

8.1. Result 1: [Upper and lower bounds on regret incurred at step t in terms of gaps Δ_{e,e^*} and the events $G_{e,e^*,t}$ conditioned on history]

Theorem 1

$$\mathbb{E}[R(A_t, w_t) | \mathcal{H}_t] \leq \sum_{e=K+1}^L \sum_{e^*=1}^K \Delta_{e,e^*} \mathbb{E}[\mathbf{1}\{G_{e,e^*,t}\} | \mathcal{H}_t]$$

$$\mathbb{E}[R(A_t, w_t) | \mathcal{H}_t] \geq \alpha \sum_{e=K+1}^L \sum_{e^*=1}^K \Delta_{e,e^*} \mathbb{E}[\mathbf{1}\{G_{e,e^*,t}\} | \mathcal{H}_t]$$

where $\alpha = (1 - \bar{w}(1))^{K-1}$, and $\bar{w}(1)$ is the attraction probability of the most attractive item.

8.2. Result 2: [Gap- dependent upper bounds on the regret of CascadeUCB1 and CascadeKL-UCB]

Theorem 2

The expected T -step regret of CascadeUCB1 is bounded as:

$$R(T) \leq \sum_{e=K+1}^L \frac{12}{\Delta_{e,K}} \log T + \frac{\pi^2}{3} L.$$

Theorem 3

For any $\varepsilon > 0$, the expected T -step regret of CascadeKL-UCB is bounded as:

$$R(T) \leq \sum_{e=K+1}^L \frac{(1 + \varepsilon) \Delta_{e,K} (1 + \log(1/\Delta_{e,K}))}{D_{\text{KL}}(\bar{w}(e) \parallel \bar{w}(K))} (\log T + 3 \log \log T) + C,$$

where $C = KL \frac{C_2(\varepsilon)}{\beta(\varepsilon)} + 7K \log \log T$, and the constants $C_2(\varepsilon)$ and $\beta(\varepsilon)$ are defined in proof.

8.3. Result 3: [Lower bound on regret for consistent algorithms on $B_{\text{easy}}(L, K, p, \Delta)$]

Theorem 4 For any cascading bandit $B_{\text{easy}}(L, K, p, \Delta)$ and any consistent algorithm, we have:

$$\liminf_{T \rightarrow \infty} \frac{R(T)}{\log T} \geq \frac{(L - K) \Delta (1 - p)^{K-1}}{D_{\text{KL}}(p - \Delta \parallel p)}.$$

9. Comparison of Our Implementation with Results reported in the Paper

For simplicity we ran simulations on $B_{\text{easy}}(L, K, 0.2, \Delta)$ instances, we will vary L , K and Δ . Remember that in pseudo-code of both algorithms we never introduced the notion of ascending or descending UCB. We can order the items in the recommended list A_t in two ways: either in descending order of UCBs or in ascending order of UCBs. The following few tables compare the regret incurred by CascadeUCB1 and CascadeKL-UCB algorithms for different values of Δ , L and K .

Table 1: n -step regret ($T = 10^5$, $n = 20$ runs, mean $\pm$ SEM($\sigma/\sqrt{n}$)) for descending UCB.

L	K	Δ	CascadeUCB1		CascadeKL-UCB	
			Paper	Ours	Paper	Ours
16	2	0.15	1290.1 $\pm$ 11.3	1286.0 $\pm$ 7.6	357.9 $\pm$ 5.5	341.7 $\pm$ 4.7
16	4	0.15	986.8 $\pm$ 10.8	1001.4 $\pm$ 7.1	275.1 $\pm$ 5.8	269.0 $\pm$ 5.6
16	8	0.15	574.8 $\pm$ 7.9	545.1 $\pm$ 4.0	149.1 $\pm$ 3.2	151.5 $\pm$ 2.8
32	2	0.15	2695.9 $\pm$ 19.8	2711.9 $\pm$ 13.0	761.2 $\pm$ 10.4	753.6 $\pm$ 10.6
32	4	0.15	2256.8 $\pm$ 12.8	2255.1 $\pm$ 11.5	633.2 $\pm$ 7.0	605.9 $\pm$ 6.9
32	8	0.15	1581.0 $\pm$ 20.3	1534.1 $\pm$ 6.3	435.4 $\pm$ 5.7	423.5 $\pm$ 5.7
16	2	0.075	2077.0 $\pm$ 32.9	2084.6 $\pm$ 23.5	766.0 $\pm$ 18.0	745.6 $\pm$ 10.5
16	4	0.075	1520.4 $\pm$ 23.4	1497.1 $\pm$ 21.7	538.5 $\pm$ 12.5	541.4 $\pm$ 11.3
16	8	0.075	725.4 $\pm$ 12.0	718.6 $\pm$ 9.3	321.0 $\pm$ 16.3	265.5 $\pm$ 6.9

Table 2: n -step regret ($T = 10^5$, $n = 20$ runs, mean $\pm$ SEM) for ascending UCB.

L	K	Δ	CascadeUCB1		CascadeKL-UCB	
			Paper	Ours	Paper	Ours
16	2	0.15	1160.2 ± 11.7	1123.3 ± 9.3	333.3 ± 6.1	335.0 ± 5.5
16	4	0.15	660.0 ± 8.3	651.4 ± 5.5	209.4 ± 4.4	198.8 ± 3.4
16	8	0.15	181.4 ± 3.9	176.8 ± 1.5	60.4 ± 2.0	61.3 ± 2.1
32	2	0.15	2471.6 ± 14.1	2446.9 ± 16.4	716.0 ± 7.5	694.1 ± 7.3
32	4	0.15	1615.3 ± 14.5	1577.5 ± 10.0	482.3 ± 6.7	499.5 ± 6.0
32	8	0.15	595.0 ± 7.8	583.0 ± 4.0	201.9 ± 5.8	204.4 ± 3.0
16	2	0.075	1989.8 ± 31.4	1954.6 ± 23.1	785.8 ± 12.2	719.9 ± 15.4
16	4	0.075	1239.5 ± 16.2	1145.5 ± 10.6	484.2 ± 12.5	445.9 ± 10.1
16	8	0.075	336.4 ± 10.3	315.3 ± 4.4	139.7 ± 6.6	132.2 ± 3.1

9.1. Comments on results

We would like to point out the reason we chose $p = 0.2$ is that it is close to $\frac{1}{K}$ for the maximum value of K we are using in our simulations ie. $K = 8$. Our lower bound mentioned in Theorem 8.3 is practical when no optimal item is very attractive, $p < 1/K$ (though why the paper cites $1/K$ as the threshold is probably to make practically p a function of K). In this case, the learning agent must learn K sufficiently attractive items to identify the optimal solution. This lower bound is not practical when p is close to 1, because it becomes exponentially small. In this case, other lower bounds would be more practical.

For instance, consider a problem with L items where item 1 is attractive with probability one and all other items are attractive with probability zero. The optimal list of K items in this problem can be found in $L/(2K)$ steps in expectation as follows : To search the database, the learning agent sequentially recommends L/K disjoint lists of K items. Because the single optimal item is uniformly distributed among these lists, the expected number of steps to find it is simply the average: $(L/K)/2$, which equals $L/(2K)$ steps.

Also we can see that regret incurred by ascending UCB is less than the regret incurred by descending UCB. This could probably be because in the case of ascending UCB, users would provide more feedback on low quality items which could speed up the learning process. However, with our current analysis we are not able to capture this phenomenon in the regret bounds.

10. Surprisingly Our model works even when the assumptions of the model are violated

10.1. Dynamic Bayesian Network (DBN) model

The DBN model is a more sophisticated model of user behavior than the cascade model. In the DBN model, the user has a probability of being satisfied with each item and may continue to examine the next item with some probability even after clicking on an item or may abandon the search altogether. The ranked bandits algorithm maintains separate bandit algorithms for each position in the ranked list, where each of those bandit algorithms has KL-UCB as the base algorithm. Despite being designed for a different user behavior model, our algorithms still

perform well in simulations where user interactions are generated according to the DBN model, demonstrating their robustness to violations of the cascade model assumptions.

10.2. Mathematical formulation of the DBN model

In the DBN extension of the cascade model, a user sequentially examines a recommended list of K items, denoted as $A = (a_1, \dots, a_K)$. The model is parameterized by three key variables:

- $\rho \in [0, 1]^E$: The **attraction probabilities**. $\rho(a_k)$ is the probability that item a_k attracts the user upon examination.
- $\nu \in [0, 1]^E$: The **satisfaction probabilities**. $\nu(a_k)$ is the probability that the user is satisfied by item a_k given that they clicked on it.
- $\gamma \in (0, 1]$: The **persistence** of the user. If the user examines an item and is not satisfied, γ is the probability that they will continue scrolling to examine the next item a_{k+1} .

Let $\bar{w}(e)$ be the joint probability that item e satisfies the user after being examined, defined as:

$$\bar{w}(e) = \rho(e)\nu(e)$$

The overall objective function, which represents the probability that at least one item in the recommended list A satisfies the user, is formulated as:

$$P(\text{Satisfied} \mid A) = \sum_{k=1}^K \gamma^{k-1} \bar{w}(a_k) \prod_{i=1}^{k-1} (1 - \bar{w}(a_i)) \quad (1)$$

We are interested in finding the optimal list of items A^* that maximizes this objective function. It turns out that this occurs when we recommend the K items with the largest values of $\bar{w}(e)$, which is the same criterion as in the original cascade model except this time order matters. The proof of this can be done using exchange argument as suppose we have $\bar{w}(a_i) \leq \bar{w}(a_{i+1})$ then if we exchange the item i with $i + 1$, the objective either increases or remains same. Also if we replace the last recommended item with some not recommended item having higher mean that also increases the objective.

10.3. Performance of CascadeUCB1, CascadeKL-UCB and RankedKL-UCB assuming DBN model of user interaction (The aim of this experiment was to highlight that despite having same base oracle KL-UCB, CascadeKL-UCB and RankedKL-UCB (probably this was academic benchmark during 2010s) have different performance characteristics suggesting better stastical efficiency)

Remark 2. The following two results are from experiments conducted with DBN interaction having parameters : $L = 16, K = 4, \rho(e) = 0.2 \forall 1 \leq e \leq K, \rho(e) = 0.05 \forall e > K, \gamma$ and $\nu(e)$ are stated in figure itself. γ and $\nu(e)$ are item independent.

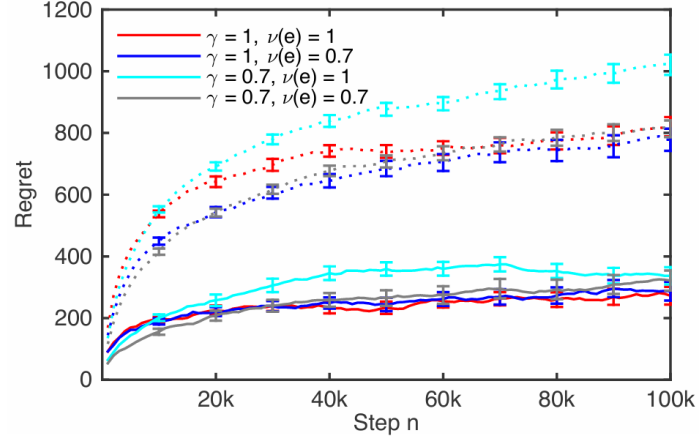


Figure 1. The n -step regret of CascadeKL-UCB (solid lines) and RankedKL-UCB (dotted lines) in the DBN model in Section 5.3.

Figure 1: above experiment result is taken from paper.

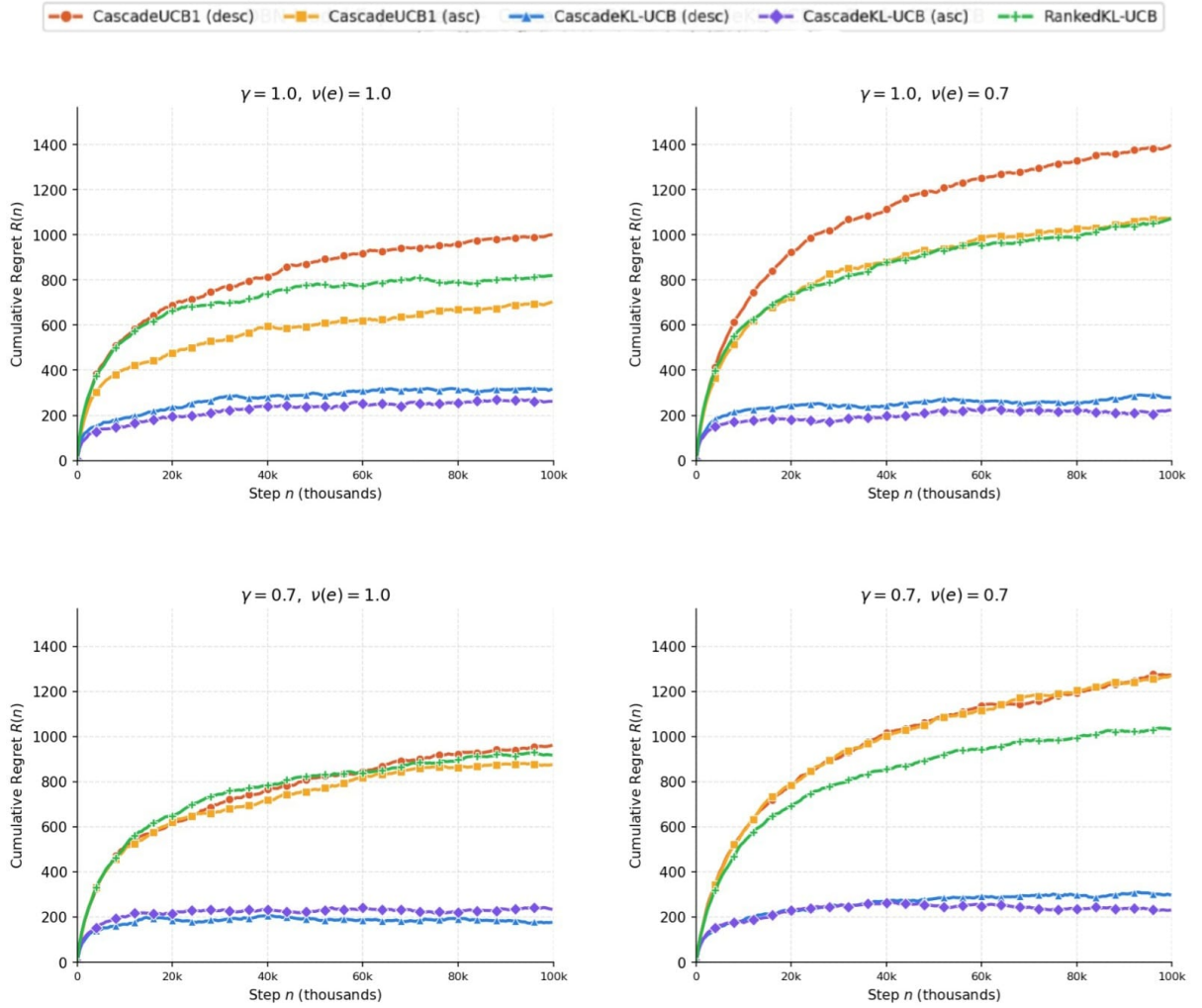


Figure 2: result is from our implementation of the same experiment

11. Further Directions

Future work will focus on expanding the theoretical foundations, understanding algorithmic behavior under varying conditions, and scaling the framework for broader applications:

Theoretical Foundations

- **Broader Lower Bounds:** Establish asymptotic lower bounds on regret for general cascading bandit instances (expanding beyond standard B_{easy} instances) and analyze the performance limits of non-consistent algorithms.
- **Gap-Free Upper Bounds:** Derive problem-independent (gap-free) regret upper bounds for both the CascadeUCB1 and CascadeKL-UCB algorithms.

Algorithmic Mechanics and Robustness

- **Theoretical Impact of Item Ordering:** Incorporate the presentation order of recommended items into the regret analysis to mathematically explain why ascending UCBs empirically yield lower regret than descending UCBs.
- **Robustness to Model Misspecification:** Formally prove the theoretical guarantees behind the empirical success of cascading bandits when applied to out-of-distribution user interaction models, such as Dynamic Bayesian Networks (DBNs).

Scalability and Broader Applications

- **Scalability to Large Action Spaces:** Address the linear dependence on the ground set size (L) in the current regret bounds, developing methods suitable for high-dimensional settings common in modern real-world applications.
- **Generalization to Complex Systems:** Extend the cascading bandit framework to more complex sequential problems, such as learning optimal routing paths in computer networks subject to unknown, probabilistic connection failures.

12. Proofs of the Main Results

Remark 3. The proofs presented here are more detailed than those in the original paper. In particular, we fill in several intermediate steps and arguments that were omitted or only briefly sketched in the paper.

We will use throughout this section the following notation: $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{H}_t]$ where $\mathcal{H}_t = (A_1, C_1, \dots, A_{t-1}, C_{t-1}, A_t)$ is the history of recommended lists and observed clicks before user clicks at time t .

12.1. Proof of Theorem 1

Lemma 12.1. Let $A = (a_1, \dots, a_K)$ and $B = (b_1, \dots, b_K)$ be any two lists of K items from $\Pi_K(E)$ such that $a_i = b_j$ only if $i = j$. Let $w \sim P$ as defined in Definition 6.1. Then:

$$\mathbb{E} \left[\prod_{k=1}^K w(a_k) \right] - \mathbb{E} \left[\prod_{k=1}^K w(b_k) \right] = \sum_{k=1}^K \mathbb{E} \left[\prod_{i=1}^{k-1} w(a_i) \right] \cdot \mathbb{E} [w(a_k) - w(b_k)] \cdot \left(\prod_{j=k+1}^K \mathbb{E} [w(b_j)] \right)$$

Proof. First, we prove that:

$$\prod_{k=1}^K w(a_k) - \prod_{k=1}^K w(b_k) = \sum_{k=1}^K \left(\prod_{i=1}^{k-1} w(a_i) \right) (w(a_k) - w(b_k)) \left(\prod_{j=k+1}^K w(b_j) \right)$$

holds for any $w \in \{0, 1\}^L$. The proof is by induction on K . The claim holds obviously for $K = 1$. Now suppose that the claim holds for any $A, B \in \Pi_{K-1}(E)$. Let $A, B \in \Pi_K(E)$. Then:

$$\begin{aligned} \prod_{k=1}^K w(a_k) - \prod_{k=1}^K w(b_k) &= \prod_{k=1}^K w(a_k) - w(b_K) \prod_{k=1}^{K-1} w(a_k) + w(b_K) \prod_{k=1}^{K-1} w(a_k) - \prod_{k=1}^K w(b_k) \\ &= (w(a_K) - w(b_K)) \prod_{k=1}^{K-1} w(a_k) + w(b_K) \left[\prod_{k=1}^{K-1} w(a_k) - \prod_{k=1}^{K-1} w(b_k) \right] \\ &= (w(a_K) - w(b_K)) \prod_{k=1}^{K-1} w(a_k) + \sum_{k=1}^{K-1} \left(\prod_{i=1}^{k-1} w(a_i) \right) (w(a_k) - w(b_k)) \left(\prod_{j=k+1}^K w(b_j) \right) \\ &= \sum_{k=1}^K \left(\prod_{i=1}^{k-1} w(a_i) \right) (w(a_k) - w(b_k)) \left(\prod_{j=k+1}^K w(b_j) \right). \end{aligned}$$

The third equality is by our induction hypothesis. Finally, note that $w(a_i)$ and $w(b_j)$ are independent for $i \neq j$ due to form of P and assumption in lemma. Therefore, we can decompose the expectation of the product as a product of expectations, and our claim follows. $\square$

Proof.

$$\mathbb{E}_t[R(A_t, w_t)] = \mathbb{E}_t \left[\prod_{k=1}^K (1 - w_t(a_k^t)) - \prod_{k=1}^K (1 - w_t(a_k^*)) \right] \quad \text{where } a_k^* = \pi_t(k)$$

Now we can apply the lemma 12.1 with $a_k = 1 - w_t(a_k^t)$ and $b_k = 1 - w_t(a_k^*)$ to get:

$$\begin{aligned} \mathbb{E}_t[R(A_t, w_t)] &= \sum_{k=1}^K \mathbb{E}_t \left[\prod_{i=1}^{k-1} (1 - w_t(a_i^t)) \right] \cdot \mathbb{E}_t [w_t(a_k^*) - w_t(a_k^t)] \cdot \left(\prod_{j=k+1}^K \mathbb{E}_t [1 - w_t(a_j^*)] \right) \\ &= \sum_{k=1}^K \mathbb{E}_t \left[\mathbf{1}_{\{G_{a_k^t, a_k^*, t}\}} \right] \cdot \Delta_{a_k^t, a_k^*} \cdot \left(\prod_{j=k+1}^K (1 - \bar{w}(a_j^*)) \right) \end{aligned}$$

$$\text{Now use the fact that } 1 \geq \prod_{j=k+1}^K (1 - \bar{w}(a_j^*)) \geq (1 - \bar{w}(1))^{K-1} = \alpha$$

$$\text{and } \sum_{k=1}^K \mathbb{E}_t \left[\mathbf{1}_{\{G_{a_k^t, a_k^*, t}\}} \right] \cdot \Delta_{a_k^t, a_k^*} = \sum_{e=K+1}^L \sum_{e^*=1}^K \Delta_{e, e^*} \mathbb{E}_t \left[\mathbf{1}_{\{G_{e, e^*, t}\}} \right]$$

The last equality holds due to conditioning on $\mathcal{H}_t$ ie. A_t as only those $\mathbf{1}_{G_{e, e^*, t}}$ will survive for which there exists k such that $e = a_k^t$ and $e^* = a_k^*$ and $\Delta_{e, e^*} \neq 0$. $\square$

12.2. Proof of Theorem 2

Lemma 12.2. *Let $\Delta_1 \geq \dots \geq \Delta_K$ be a sequence of K positive numbers. Then:*

$$\left[\Delta_1 \frac{1}{\Delta_1^2} + \sum_{k=2}^K \Delta_k \left(\frac{1}{\Delta_k^2} - \frac{1}{\Delta_{k-1}^2} \right) \right] \leq \frac{2}{\Delta_K}.$$

Proof. First, we note that:

$$\Delta_1 \frac{1}{\Delta_1^2} + \sum_{k=2}^K \Delta_k \left(\frac{1}{\Delta_k^2} - \frac{1}{\Delta_{k-1}^2} \right) = \sum_{k=1}^{K-1} \frac{\Delta_k - \Delta_{k+1}}{\Delta_k^2} + \frac{1}{\Delta_K}.$$

Second, by our assumption, $\Delta_k \geq \Delta_{k+1}$ for all $k < K$. Therefore:

$$\begin{aligned} \sum_{k=1}^{K-1} \frac{\Delta_k - \Delta_{k+1}}{\Delta_k^2} + \frac{1}{\Delta_K} &\leq \sum_{k=1}^{K-1} \frac{\Delta_k - \Delta_{k+1}}{\Delta_k \Delta_{k+1}} + \frac{1}{\Delta_K} \\ &= \sum_{k=1}^{K-1} \left(\frac{1}{\Delta_{k+1}} - \frac{1}{\Delta_k} \right) + \frac{1}{\Delta_K} \\ &= \frac{2}{\Delta_K} - \frac{1}{\Delta_1} < \frac{2}{\Delta_K}. \end{aligned}$$

Proof. Let $R_t = R(A_t, w_t)$ be the regret of the learning algorithm at time t , where A_t is the recommended list at time t and w_t are the weights of items at time t . Let

$$\mathcal{E}_t = \{ \exists e \in E \text{ s.t. } |\bar{w}(e) - \hat{w}_{N_{t-1}(e)}(e)| \geq c_{t-1, N_{t-1}(e)} \}$$

be the event that $\bar{w}(e)$ is not in the high-probability confidence interval around $\hat{w}_{N_{t-1}(e)}(e)$ for some e at time t where $c_{t,s} = \sqrt{\frac{1.5 \log t}{s}}$ and let $\bar{\mathcal{E}}_t$ be its complement. Then we can decompose the regret as:

$$R(n) = \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\mathcal{E}_t\} R_t \right] + \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\bar{\mathcal{E}}_t\} R_t \right]. \quad (10)$$

We now bound both terms.

The first term is small because the confidence intervals hold with high probability. By Hoeffding's inequality, for any e, s, t :

$$\mathbb{P}(|\bar{w}(e) - \hat{w}_s(e)| \geq c_{t,s}) \leq 2 \exp(-2sc_{t,s}^2) = 2 \exp(-2s \frac{3 \log t}{2s}) \leq 2 \exp(-3 \log t).$$

Thus,

$$\mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\mathcal{E}_t\} \right] = \sum_{t=1}^T \mathbb{P}(\mathcal{E}_t) \leq \sum_{e \in E} \sum_{t=1}^T \sum_{s=1}^t \mathbb{P}(|\bar{w}(e) - \hat{w}_s(e)| \geq c_{t,s}) \leq 2 \sum_{e \in E} \sum_{t=1}^T t^{-2} \leq \frac{\pi^2}{3} L. \quad (2)$$

Here we applied union bound on items e and as well as $N_{t-1}(e)$.

Since $R_t \leq 1$, we get $\mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\mathcal{E}_t\} R_t \right] \leq \frac{\pi^2}{3} L$.

$$\mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\bar{\mathcal{E}}_t\} R_t \right] = \sum_{t=1}^T \mathbb{E}_{\mathcal{H}_t} [\mathbf{1}\{\bar{\mathcal{E}}_t\} \mathbb{E}_t[R_t]] \leq \sum_{e=K+1}^L \mathbb{E} \left[\sum_{e^*=1}^K \sum_{t=1}^T \Delta_{e,e^*} \mathbf{1}\{\bar{\mathcal{E}}_t, G_{e,e^*,t}\} \right].$$

We used the fact that conditioned on $\mathcal{H}_t$, $\mathbf{1}\{\bar{\mathcal{E}}_t\}$ is just a constant. We did not show how $\mathbb{E}_{\mathcal{H}_t}$ changes to $\mathbb{E}$ in last inequality but it is just reverse tower rule of expectation.

Now fix a suboptimal item e . All the following arguments will hold for this e . When $\bar{\mathcal{E}}_t$ holds $|\bar{w}(e) - \hat{w}_{N_{t-1}(e)}(e)| < c_{t-1, N_{t-1}(e)}$. Also, when $G_{e,e^*,t}$ holds, $U_t(e) \geq U_t(e^*)$. Hence $\bar{w}(e) + 2c_{t-1, N_{t-1}(e)} \geq U_t(e) \geq U_t(e^*) \geq \bar{w}(e^*) \implies 2c_{t-1, N_{t-1}(e)} \geq \Delta_{e,e^*}$.

Using $c_{T,s} \geq c_{t-1,s}$ we get $N_{t-1}(e) \leq \tau_{e,e^*}$, where $\tau_{e,e^*} = \frac{6 \log T}{\Delta_{e,e^*}^2}$.

Thus $\sum_{e^*=1}^K \sum_{t=1}^T \Delta_{e,e^*} \mathbf{1}\{\bar{\mathcal{E}}_t, G_{e,e^*,t}\} \leq \sum_{e^*=1}^K \Delta_{e,e^*} \sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) \leq \tau_{e,e^*}, G_{e,e^*,t}\}$ since event $\bar{\mathcal{E}}_t$ implies event $N_{t-1}(e) \leq \tau_{e,e^*}$.

$N_{t-1}(e)$ increases iff $G_{e,e^*,t}$ holds for exactly one e^* at time t conditioned on $\mathcal{H}_t$. We use this to get the following

$$\begin{aligned} M_{e,e^*} &= \sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) \leq \tau_{e,e^*}, G_{e,e^*,t}\} \\ \implies \sum_{e^* \in S \subseteq [K]} M_{e,e^*} &= \sum_{e^* \in S \subseteq [K]} \sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) \leq \tau_{e,e^*}, G_{e,e^*,t}\} \\ &\leq \sum_{e^* \in S \subseteq [K]} \sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) \leq \tau_{e, \max(S)}, G_{e,e^*,t}\} \quad \text{as } \tau_{e,i} \text{ increases with } i \\ &\leq \sum_{e^* \in [K]} \sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) \leq \tau_{e, \max(S)}, G_{e,e^*,t}\} \\ &\leq \sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) \leq \tau_{e, \max(S)}, \exists! e^* \in [K] : G_{e,e^*,t}\} \leq \tau_{e, \max(S)}. \end{aligned}$$

Hence $\sum_{e^*=1}^K \Delta_{e,e^*} M_{e,e^*} \leq \max \left\{ \sum_{e^*=1}^K \Delta_{e,e^*} m_{e,e^*} : 0 \leq m_{e,e^*} \leq \tau_{e,e^*}, \sum m_{e,e^*} \leq \tau_{e,K} \right\}$. Be careful that $m_{e,e^*} \neq M_{e,e^*}$. Since $\Delta_{e,1} \geq \dots \geq \Delta_{e,K}$, the optimal solution is $m_{e,1}^* = \tau_{e,1}$, $m_{e,2}^* = \tau_{e,2} - \tau_{e,1}, \dots$

$$\implies \sum_{e^*=1}^K \Delta_{e,e^*} M_{e,e^*} \leq \left[\Delta_{e,1} \frac{1}{\Delta_{e,1}^2} + \sum_{e^*=2}^K \Delta_{e,e^*} \left(\frac{1}{\Delta_{e,e^*}^2} - \frac{1}{\Delta_{e,e^*-1}^2} \right) \right] 6 \log T.$$

By Lemma 12.2 this is bounded by:

$$\frac{12}{\Delta_{e,K}} \log T.$$

Rest is just summing over all suboptimal items e and adding the bound on the first term in equation 2 to get the final regret bound. $\square$

12.3. Proof Sketch of Theorem 3

Lemma 12.3. Let $p_1 \geq \dots \geq p_K > p$ be $K+1$ probabilities and $\Delta_k = p_k - p$ for $1 \leq k \leq K$. Then:

$$\frac{\Delta_1}{D_{\text{KL}}(p \| p_1)} + \sum_{k=2}^K \Delta_k \left(\frac{1}{D_{\text{KL}}(p \| p_k)} - \frac{1}{D_{\text{KL}}(p \| p_{k-1})} \right) \leq \frac{\Delta_K (1 + \log(1/\Delta_K))}{D_{\text{KL}}(p \| p_K)}.$$

Proof.

$$\frac{\Delta_1}{D_{\text{KL}}(p \| p_1)} + \sum_{k=2}^K \Delta_k \left(\frac{1}{D_{\text{KL}}(p \| p_k)} - \frac{1}{D_{\text{KL}}(p \| p_{k-1})} \right) = \sum_{k=1}^{K-1} \frac{\Delta_k - \Delta_{k+1}}{D_{\text{KL}}(p \| p_k)} + \frac{\Delta_K}{D_{\text{KL}}(p \| p_K)}.$$

The summation over k can be bounded from above by a definite integral:

$$\sum_{k=1}^{K-1} \frac{\Delta_k - \Delta_{k+1}}{D_{\text{KL}}(p \| p_k)} = \sum_{k=1}^{K-1} \frac{\Delta_k - \Delta_{k+1}}{D_{\text{KL}}(p \| p + \Delta_k)} \leq \int_{\Delta_K}^{\Delta_1} \frac{1}{D_{\text{KL}}(p \| p + x)} dx \leq \int_{\Delta_K}^1 \frac{1}{D_{\text{KL}}(p \| p + x)} dx,$$

where the first inequality follows from the fact that $1/D_{\text{KL}}(p \| p + x)$ decreases on $x \geq 0$.

To the best of our knowledge, the integral of $1/D_{\text{KL}}(p \| p + x)$ over x does not have a simple analytic solution. Therefore, we integrate an upper bound on $1/D_{\text{KL}}(p \| p + x)$ which does. In particular, note that for any $x \geq \Delta_K$ because $D_{\text{KL}}(p \| p + x)$ is convex, increasing in $x \geq 0$, and its minimum is attained at $x = 0$, let $\Delta_K = (1 - \lambda) \cdot 0 + \lambda \cdot x$, $D_{\text{KL}}(p \| p + \Delta_K) \leq (1 - \lambda) \cdot D_{\text{KL}}(p \| p) + \lambda \cdot D_{\text{KL}}(p \| p + x)$

$$D_{\text{KL}}(p \| p + x) \geq \frac{D_{\text{KL}}(p \| p + \Delta_K)}{\Delta_K} x = \frac{D_{\text{KL}}(p \| p_K)}{\Delta_K} x,$$

Therefore:

$$\int_{\Delta_K}^1 \frac{1}{D_{\text{KL}}(p \| p + x)} dx \leq \frac{\Delta_K}{D_{\text{KL}}(p \| p_K)} \int_{\Delta_K}^1 \frac{1}{x} dx = \frac{\Delta_K}{D_{\text{KL}}(p \| p_K)} \log(1/\Delta_K).$$

Finally, we chain all inequalities and get the result. $\square$

Proof. Define $\mathcal{E}_t$ as the event that for some optimal item its upper confidence bound is less than its true mean, mathematically $\mathcal{E}_t = \{\exists e^* \in [K] : U_t(e^*) < \bar{w}(e^*)\}$. Then we can decompose the regret as: $R(T) = \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\mathcal{E}_t\} R_t \right] + \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\bar{\mathcal{E}}_t\} R_t \right]$. We will skip the proof as it involves martingale theory and directly state that the first term on RHS is bounded by $7K \log \log T$.

$$\mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{\bar{\mathcal{E}}_t\} R_t \right] = \sum_{t=1}^T \mathbb{E}_{\mathcal{H}_t} [\mathbf{1}\{\bar{\mathcal{E}}_t\} \mathbb{E}_t[R_t]] \leq \sum_{e=K+1}^L \mathbb{E} \left[\sum_{e^*=1}^K \sum_{t=1}^T \Delta_{e,e^*} \mathbf{1}\{\bar{\mathcal{E}}_t, G_{e,e^*,t}\} \right].$$

Here we used the fact that conditioned on $\mathcal{H}_t$, $\mathbf{1}\{\bar{\mathcal{E}}_t\}$ is just a constant. We did not show how $\mathbb{E}_{\mathcal{H}_t}$ changes to $\mathbb{E}$ in last inequality but it is just reverse tower rule of expectation. Now note that for any suboptimal item e and $\tau_{e,e^*} > 0$ (Note this is different from τ_{e,e^*} defined in the proof of Theorem 2) we have:

$$\begin{aligned} \mathbb{E} \left[\sum_{e^*=1}^K \sum_{t=1}^T \Delta_{e,e^*} \mathbf{1}\{\bar{\mathcal{E}}_t, G_{e,e^*,t}\} \right] &\leq \mathbb{E} \left[\sum_{e^*=1}^K \sum_{t=1}^T \Delta_{e,e^*} \mathbf{1}\{N_{t-1}(e) \leq \tau_{e,e^*}, G_{e,e^*,t}\} \right] \\ &\quad + \sum_{e^*=1}^K \Delta_{e,e^*} \mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) > \tau_{e,e^*}, \bar{\mathcal{E}}_t, G_{e,e^*,t}\} \right]. \end{aligned} \quad (13)$$

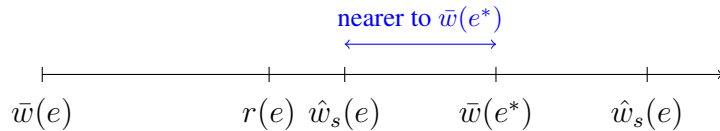
Let:

$$\tau_{e,e^*} = \frac{1 + \epsilon}{D_{\text{KL}}(\bar{w}(e) \| \bar{w}(e^*))} (\log T + 3 \log \log T).$$

Observe that when $\bar{\mathcal{E}}_t$ and $G_{e,e^*,t}$ hold, we have $U_t(e) \geq U_t(e^*) \geq \bar{w}(e^*)$. The way we construct $U_t(e)$, we have $N_{t-1}(e)d(\hat{w}_{N_{t-1}(e)}, U_t(e)) \leq \log t + 3 \log \log t \leq \log T + 3 \log \log T$ which implies $N_{t-1}(e)d^+(\hat{w}_{N_{t-1}(e)}, \bar{w}(e^*)) \leq \log T + 3 \log \log T$.

$$\begin{aligned}
& \sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) > \tau_{e,e^*}, \bar{\mathcal{E}}_t, G_{e,e^*,t}\} \\
& \leq \sum_{t=1}^T \mathbf{1}\{G_{e,e^*,t}, N_{t-1}(e)d^+(\hat{w}_{N_{t-1}(e)}, \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \\
& \leq \sum_{t=1}^T \sum_{s=\mathcal{K}}^t \mathbf{1}\{G_{e,e^*,t}, N_{t-1}(e) = s, sd^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \quad \text{where } \mathcal{K} = \lceil \tau_{e,e^*} \rceil \\
& \leq \sum_{t=1}^T \sum_{s=\mathcal{K}}^t \mathbf{1}\{G_{e,e^*,t}, N_{t-1}(e) = s\} \mathbf{1}\{sd^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \\
& \leq \sum_{s=\mathcal{K}}^T \sum_{t=s}^T \mathbf{1}\{G_{e,e^*,t}, N_{t-1}(e) = s\} \mathbf{1}\{sd^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \\
& \leq \sum_{s=\mathcal{K}}^T \mathbf{1}\{sd^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \sum_{t=s}^T \mathbf{1}\{G_{e,e^*,t}, N_{t-1}(e) = s\} \\
& \leq \sum_{s=\mathcal{K}}^T \mathbf{1}\{sd^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \\
& \leq \sum_{s=\mathcal{K}}^T \mathbf{1}\{\mathcal{K}d^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \\
& \leq \sum_{s=\mathcal{K}}^T \mathbf{1}\{\tau_{e,e^*}d^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \log T + 3 \log \log T\} \\
& \leq \sum_{s=\mathcal{K}}^T \mathbf{1}\left\{d^+(\hat{w}_s(e), \bar{w}(e^*)) \leq \frac{d(\bar{w}(e), \bar{w}(e^*))}{(1+\epsilon)}\right\}
\end{aligned}$$

holds for any suboptimal e and optimal e^* . Now let $\frac{d(\bar{w}(e), \bar{w}(e^*))}{1+\epsilon} = d(r(\epsilon), \bar{w}(e^*))$ for some $r(\epsilon) \in [\bar{w}(e), \bar{w}(e^*)]$.



$\mathbb{E} \left[\sum_{t=1}^T \mathbf{1}\{N_{t-1}(e) > \tau_{e,e^*}, \bar{\mathcal{E}}_t, G_{e,e^*,t}\} \right] \leq \sum_{s=\mathcal{K}}^T \mathbb{P}(d^+(\hat{w}_s(e), \bar{w}(e^*)) \leq d(r(\epsilon), \bar{w}(e^*))) \leq \sum_{s=\mathcal{K}}^\infty \mathbb{P}(d^+(\hat{w}_s(e), \bar{w}(e^*)) \leq d(r(\epsilon), \bar{w}(e^*))) \leq \sum_{s=\mathcal{K}}^\infty \mathbb{P}(r(\epsilon) \leq \hat{w}_s(e)) \leq \sum_{s=\mathcal{K}}^\infty \exp(-sd(r(\epsilon), \bar{w}(e))) \leq \frac{C_2(\epsilon)}{n^{\beta(\epsilon)}} \text{ where } C_2(\epsilon) = \frac{1}{1-\exp(-d(r(\epsilon), \bar{w}(e)))} \text{ and } \beta(\epsilon) = \frac{(1+\epsilon)d(r(\epsilon), \bar{w}(e))}{d(\bar{w}(e), \bar{w}(e^*))}. \text{ The last inequality follows due to following inequality for iid bernoulli random variables } X_i \text{ with mean } p, \mathbb{P}(\hat{X}_n \geq q) \leq \exp(-nd(p, q)) \text{ for any } q > p \text{ where } \hat{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$

So the second term in (13) is bounded from above by $K \frac{C_2(\epsilon)}{n^{\beta(\epsilon)}}$.

Now we bound the first term in (13). By the same argument as in the proof of Theorem 2:

$$\begin{aligned} & \sum_{e^*=1}^K \sum_{t=1}^T \Delta_{e,e^*} \mathbf{1}\{N_{t-1}(e) \leq \tau_{e,e^*}, G_{e,e^*,t}\} \\ & \leq \left[\frac{\Delta_{e,1}}{D_{\text{KL}}(\bar{w}(e) \parallel \bar{w}(1))} + \sum_{e^*=2}^K \Delta_{e,e^*} \left(\frac{1}{D_{\text{KL}}(\bar{w}(e) \parallel \bar{w}(e^*))} - \frac{1}{D_{\text{KL}}(\bar{w}(e) \parallel \bar{w}(e^* - 1))} \right) \right] (1 + \epsilon)(\log T + 3 \log \log T). \end{aligned}$$

Now using lemma 12.3 we are done. □

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